

Building the Web for Everyone

Web Accessibility 101





What Is Accessibility?

- **Making websites usable by everyone**
- ...including people with disabilities
- **Disabilities that affect web use:**
 - Visual — blindness, low vision, color blindness
 - Motor — limited hand movement, tremors
 - Cognitive — ADHD, dyslexia, memory issues
 - Auditory — deafness, hearing loss

1 in 4

US adults has a disability

Source: CDC

"You're probably already know how to design for yourself — let's learn to design for everyone."



Why Does It Matter?



Legal

ADA compliance applies to websites — real lawsuits happen against businesses with inaccessible sites.



Ethical

Everyone deserves access to information, services, and opportunities online.



Practical

Accessible sites are better for ALL users. Captions help in noisy rooms. Keyboard nav helps developers.

Universal Design Insight: Curb cuts were designed for wheelchair users — but they also help people with strollers, delivery workers, and cyclists.

The POUR Principles (WCAG)

WCAG = Web Content Accessibility Guidelines — the official rulebook for the web

P

Perceivable

Can everyone see or hear your content?

Alt text, captions, readable font sizes

O

Operable

Can everyone use and navigate your site?

Keyboard navigation, no time limits, visible focus

U

Understandable

Is your content clear to read and use?

Plain language, clear labels, consistent layouts

R

Robust

Does your site work with assistive tech?

Valid semantic HTML, works with screen readers

</> Semantic HTML: The Foundation

"Semantic" = using the right HTML tag for its **meaning**, not just its looks

Screen readers announce tag purpose out loud:

"navigation landmark"

"heading level 1"

"button"

No semantic meaning = screen reader stays silent.

X `<div
class="nav">`



✓ `<nav>`

X `<p class="big-
text">`



✓ `<h1>`

X `<div
class="btn">`



✓ `<button>`

Lab Connection: Steps 2–5: Add lang="en" (2), unique page title (3), club name in <h1> (4), and favicon (5)

Images & Alt Text

Screen readers read the alt attribute aloud — blind users rely on it to understand images.

✗ Bad

```
alt="image1.jpg"
```

File name means nothing to a screen reader user

~ Okay

```
alt="photo"
```

Tells us it's a photo, but of what?

✓ Best

```
alt="Students  
collaborating  
on laptops"
```

Describes who, what, and context

Tip: Purely decorative images should use `alt=""` so screen readers skip them.

Lab Connection: Steps 7 & 21: Add `<figure>` + `<figcaption>` to image cards (Step 7); fix missing alt text on footer logo (Step 21)



Forms & Labels

Why Labels Matter

- Labels tell people what to enter in a form.
- Screen readers read those labels aloud.
- Without labels, a user may only hear ***"edit text"*** instead of knowing what the field is for.

Remember

- Every input needs a label
- Labels should be clear
- Related options need a group label

```
<label for="email"> Email </label>
```

```
<input id="email" type="email">
```

```
<fieldset>
```

```
  <legend>Best Day?</legend>
```

```
  <input type="radio" ...
```

```
</fieldset>
```

Lab Connection: Steps 17, 19 & 20: Rename q1/q2 to descriptive names (17), add <legend> to fieldsets (19), fix email label (20)



Color & Contrast

8% of men have color blindness
— never rely on color alone to convey info

WCAG minimum contrast ratio: **4.5:1** for normal text

Ask yourself:

"Is it readable in bright sunlight?"

"Does it still work in grayscale?"

Contrast Examples

Low Contrast — Hard to Read

Fails WCAG (ratio: ~1.5:1)

Medium — Borderline

Check with axe (ratio: ~3.5:1)

High Contrast — Passes!

Passes WCAG AA (ratio: 8.5:1)



Keyboard Navigation

Some users can't use a mouse — they rely entirely on the keyboard to navigate.

Tab

Move forward
to next element

**Shift
+Tab**

Move backward
to previous

**Enter /
Space**

Activate button,
link, or checkbox

↑ ↓ ← →

Navigate within
menus & radio groups

Visible Focus Indicator

Each focused element should have a visible outline. Never remove it with `outline: none` — keyboard users need to know WHERE they are!

Lab Connection: Step 24: With NVDA on, use the Tab key to navigate interactive elements and verify visible focus on every input, link, and button

axe DevTools

Browser Extension — Free

- Automated WCAG scan in seconds
- Flags: missing alt text, bad contrast, broken labels, heading errors
- Finds ~57% of all a11y issues
- Great starting point before manual testing

NVDA Screen Reader

Free Windows App

- Hear your page as a blind user would
- ↓ Arrow: read next element
- H: jump to next heading
- Announces: links, buttons, landmarks
- Reveals gaps automated tools miss

"Automated testing finds the obvious bugs — manual testing reveals the real experience."

Lab Connection: axe DevTools: Steps 10 (club page) & 22 (registration) | NVDA: Steps 11 (club page) & 23 (registration)